

LinearSkills Academy

Class 6 - for Loop

What is a for loop?

A loop repeats a task. A for loop is used when you already know how many times to repeat it.

1. Basic for Loop

```
for i in range(5):  
    print(i)
```

Output: 0 1 2 3 4

2. range() - short definition

range() generates a sequence of numbers for the loop to use.

Syntax	Example	Output
range(stop)	range(5)	0, 1, 2, 3, 4
range(start, stop)	range(2, 6)	2, 3, 4, 5
range(start, stop, step)	range(0, 10, 2)	0, 2, 4, 6, 8
reverse	range(5, 0, -1)	5, 4, 3, 2, 1

3. Iterating a String

A string is looped character by character.

```
for letter in "Hi":  
    print(letter)
```

4. Python vs C++ for Loop

Feature	Python	C++
Loop line	for i in range(5):	for(int i=0;i<5;i++){
Counter	Handled automatically	Declared manually
Block	Colon + indentation	Curly braces { }